AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

Particulate Matter
Monitoring & Health Effects

Issue 2026-10-20 | Entry window 20 October 2026 (full day) | 19 records screened in scope

19 RECORDS IN SCOPE (58 EXCLUDED)

8 OF 10 SUBTOPIC BANDS POPULATED

2 EFFECT ESTIMATES WITH A CI

5 TIER-A RECORDS

6 RECORDS CARRIED ON METADATA ALONE

Signal of the day

Four identical PM monitors breathing the same air disagree by up to a factor of three day to day — and averaging does not fix it.

Four prototype electrostatic (corona-charging, impaction, Faraday-cup) $PM_{10}/PM_{2.5}/PM_{1.0}$ monitors ran at one Thai combined-cycle power plant for four months with 92–100 % daily recovery. Fixed station differences explained only **0.7 %** of log 24-h PM_{10} variance; the day×station interaction explained **57.1 %**. Pairwise 95 % limits of agreement spanned ratios **0.35–3.26** (PM_{10}) and 0.33–2.83 ($PM_{2.5}$); single-unit ICC 0.42, rising to 0.75 only for the four-unit mean. **Where it is weak:** preprint, no reference instrument, and no side-by-side co-location before deployment — unit drift and real micro-gradients are separated by inference, not measurement. **Why it matters here:** the failure mode is generic to distributed networks. Station-to-station contrasts from any low-cost array need scheduled co-location checks, or they support only pooled, time-averaged statements.

What changed today

- **Fire smoke and the kidney.** 12.5 million renal admissions, 1020 locations, six countries: per 10 $\mu g\,m^{-3}$ fire-sourced $PM_{2.5}$ (lag 0–5) RR **1.011** (1.007–1.015), strongest for acute kidney injury and larger than for non-fire $PM_{2.5}$.
- **An LCS protocol for pregnancy cohorts.** PRECISE (Gambia, Kenya, Mozambique): 343 women, 3,190 person-days of wearable $PM_{2.5}$, Deming-harmonised sensors with inter-sensor R^2 >0.75 in country — precision reported, accuracy against reference not in the abstract.
- **AI inside the CTM, not after it.** ProNet embeds neural correctors in the transport equations; across 338 Chinese cities it beats forecaster-adjusted operational $PM_{2.5}$ guidance by 6.1 % and attributes errors to processes.
- **Same emissions, more PM.** CAMS reanalysis: the Indo-Gangetic Plain emits 40 % less than the North China Plain yet averages 123.6 vs 113.2 $\mu g\,m^{-3}$, with winter 'pollution-forming efficiency' nearly double under a <300 m boundary layer.
- **Held record released.** *Foehn* (arXiv 2610.25941), the generative km-scale global $PM_{2.5}$ forecaster held from 19 October, now resolves at doi.org and enters this issue.

Window continuity

The 19 October issue closed with `last_entry_date` at 19 October. This issue collects **20 October in full**, continuous with it. Built on the run that started 23:0x CDT on 20 October (04:0x UTC 21 October), after the 22:00 cut-off, so the day is complete. 21 October is due on the next run at or after 22:00 CDT; the W43 weekly (18–24 October) on Saturday 24 October.

Corpus analytics

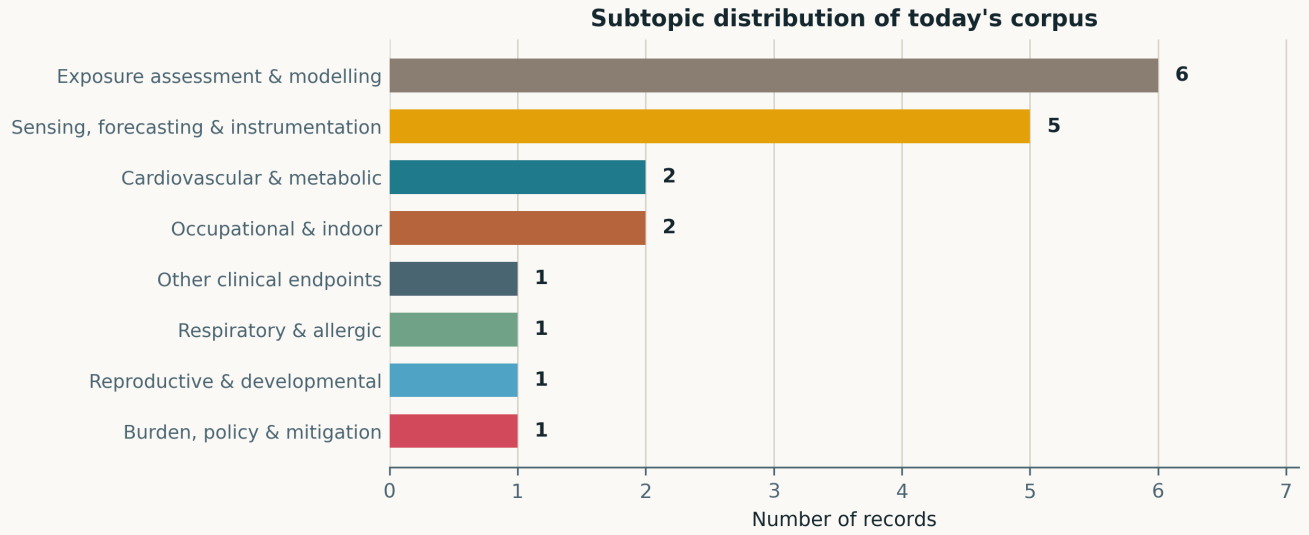


Fig. Subtopic assignment of the 19 in-scope records across eight bands. Exposure assessment (6) and sensing/forecasting (5) lead — a measurement-heavy day; cardiovascular and occupational hold two each; neuro/mental health and mechanistic toxicology are empty.

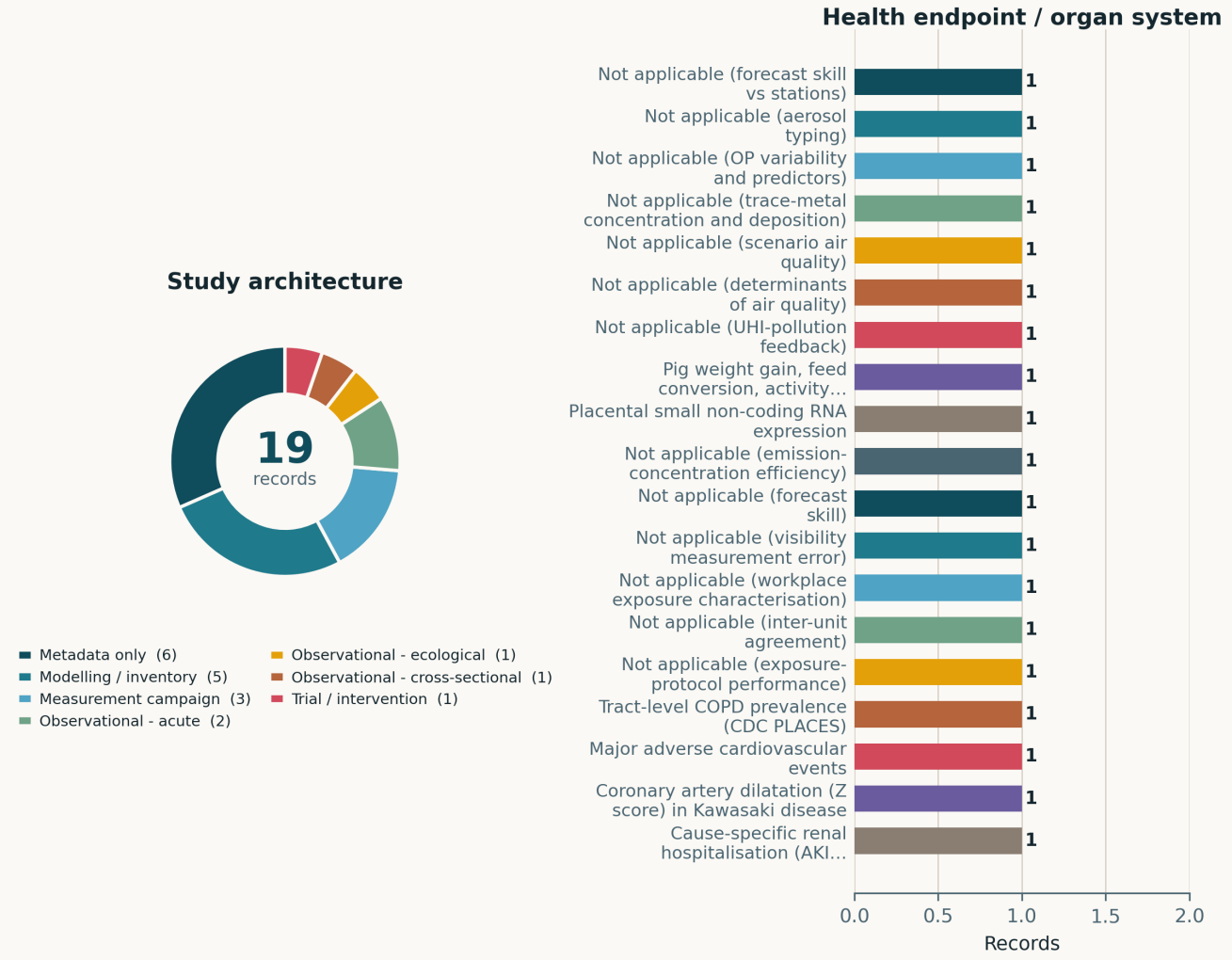


Fig. Left — architecture: modelling/inventory 5, measurement campaigns 3, two acute (case-crossover, multi-location time-series), one ecological, one cross-sectional, one non-randomised intervention; **metadata-only 6 (32 %)**, again day-0 Elsevier records. Right — endpoints: renal admissions, coronary dilatation, MACE, COPD prevalence, placental sncRNA, livestock productivity.

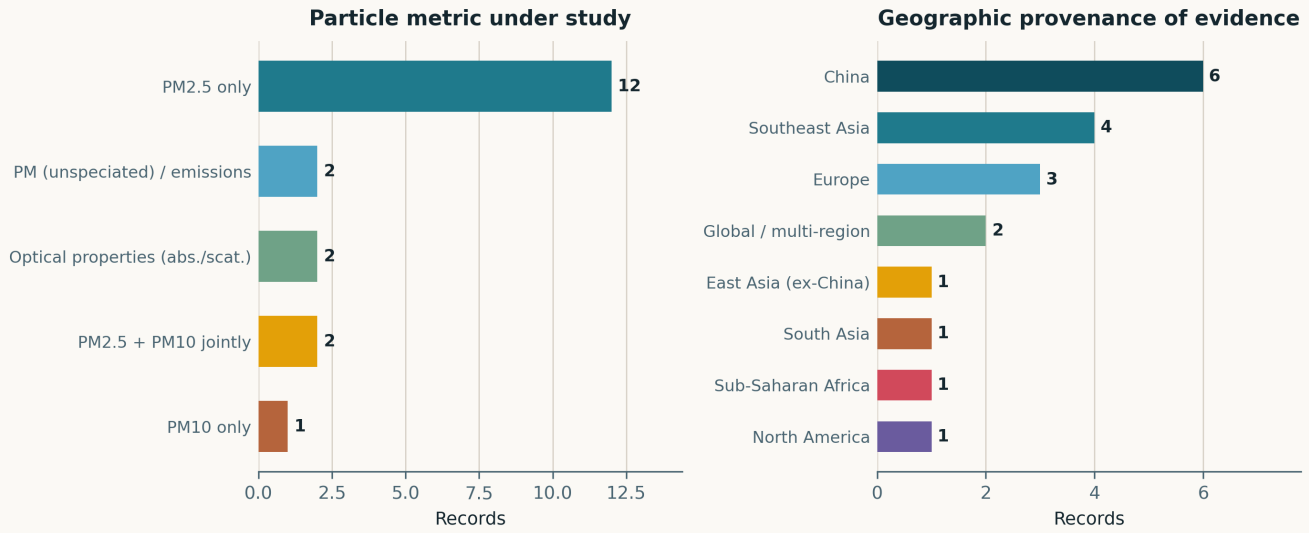


Fig. Left — particle metric: $PM_{2.5}$-only records dominate (12); two size-resolved, two optical-property (brown carbon, aerosol typing). Right — geography: China 6, Southeast Asia 4 (Thailand, Singapore), Europe 3, then sub-Saharan Africa, South Asia, Korea and the US border.

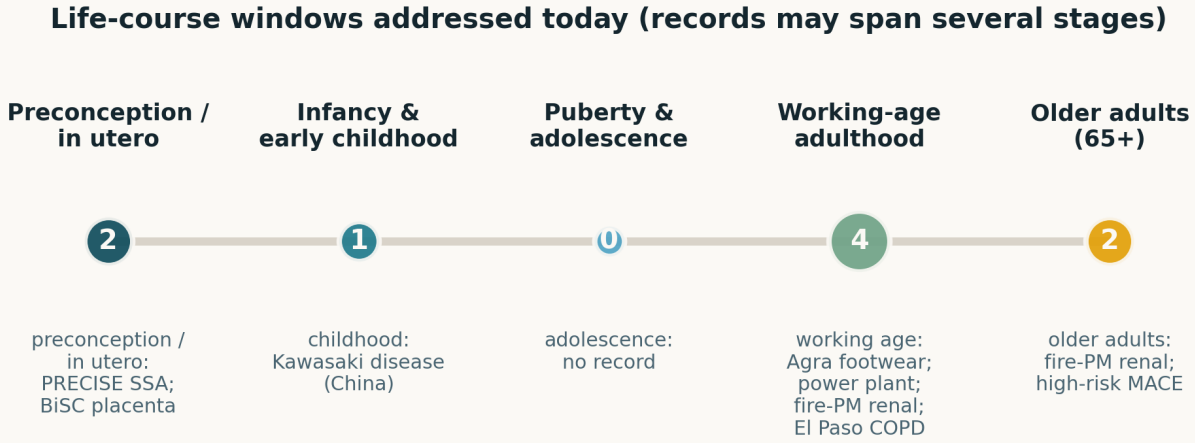


Fig. Life-course windows: working age leads (two occupational campaigns plus the adult renal and COPD records); two in utero (PRECISE, BiSC placenta), one childhood (Kawasaki); no adolescent record.

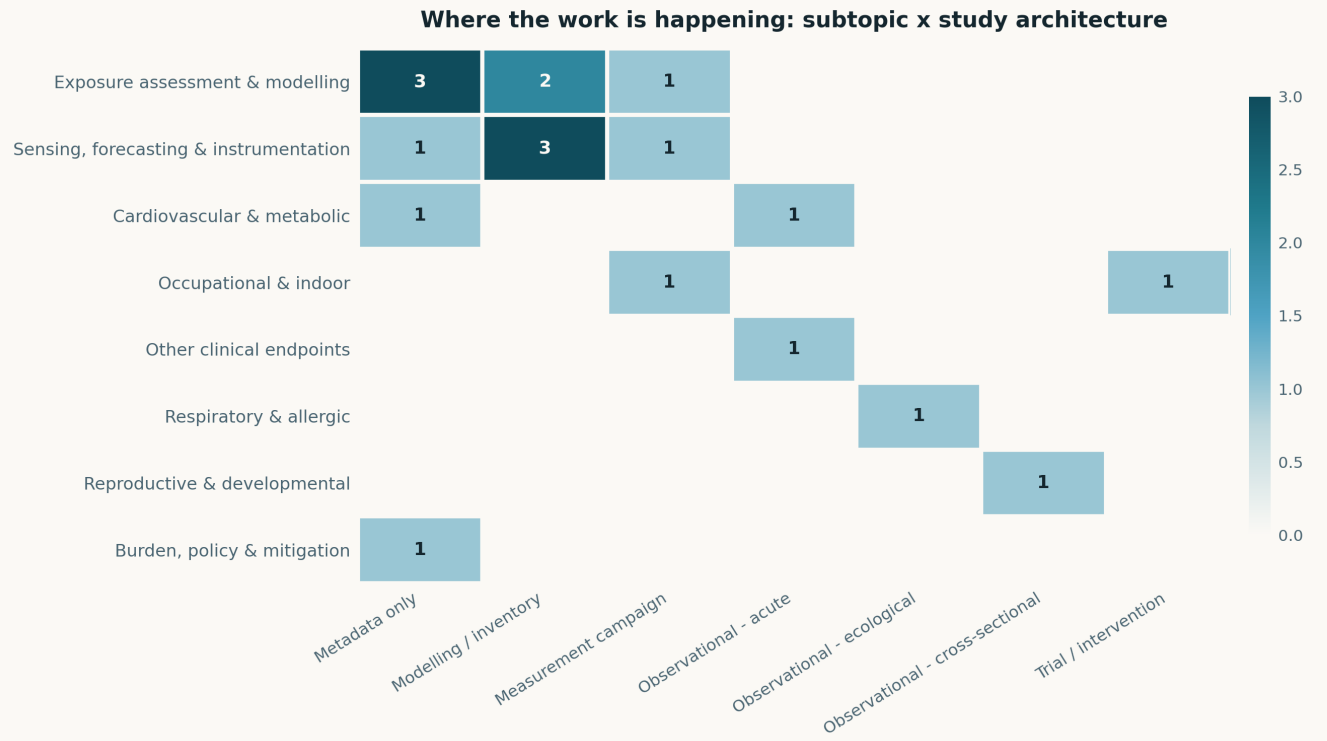


Fig. Subtopic × study architecture. Exposure assessment spreads across modelling and metadata-only cells; sensing concentrates in modelling (forecasters) and measurement (electrostatic monitors).

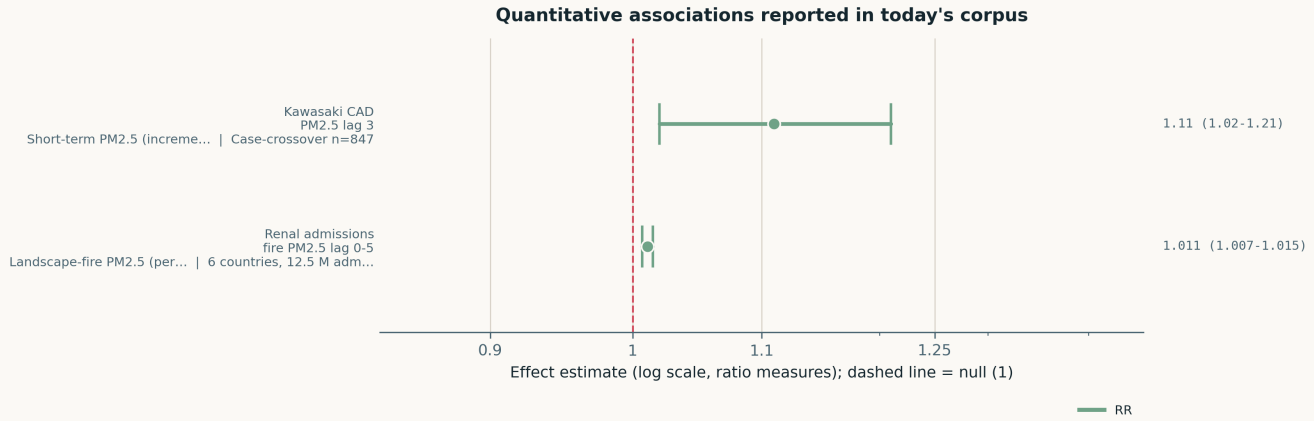


Fig. Ratio scale, two estimates on different contrasts: fire-sourced $PM_{2.5}$ and renal admissions per $10\,\mu g\,m^{-3}$ over lag 0–5 (six countries), and $PM_{2.5}$ and coronary artery dilatation in Kawasaki disease at lag 3, whose increment the abstract does not state. Not poolable.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked *metadata only* carry no recoverable abstract and assert no finding.

Cardiovascular & metabolic — 2 records

Huang $PM_{2.5}$ coronary artery dilatation Kawasaki disease case-crossover 2026 Eur J Pediatr

Association between PM(2.5) and coronary artery dilatation with Kawasaki disease in China

847 children with Kawasaki disease, 2017-2021; case-crossover with a distributed-lag Poisson GLM on coronary Z score. Risk of coronary artery dilatation rose at lags 2-4, peaking at lag 3 with RR 1.11 (1.02-1.21); the abstract does not state the $PM_{2.5}$ increment. Mediation via CRP, ALT and haemoglobin. Limitation: a case-crossover needs an acute event with a datable onset, yet CAD is detected at echocardiography whose timing follows admission and IVIG, so the referent-window logic is shaky; the 'mediators' are markers of disease severity measured after onset, and the highlights' claim of a 'population-based cohort' contradicts the design. Relevance: low - city-monitor assignment, no sensing implication.

doi: 10.1007/s00431-026-07527-9

Wongcharoen short-term $PM_{2.5}$ MACE high-risk population nationwide case-crossover 2026 BMJ Open

Associations between short-term PM(2.5) exposure and major cardiovascular events in high-risk population: a nationwide case-crossover study

METADATA ONLY. Title and authors only (Wongcharoen et al.); no abstract indexed at harvest. A nationwide time-stratified case-crossover of short-term $PM_{2.5}$ and major adverse cardiovascular events restricted to a 'high-risk population' (presumably prior CVD, diabetes or hypertension). Country, n, lag structure and effect size cannot be stated. Limitation: unverifiable beyond the title; case-crossover in a pre-selected high-risk group estimates a conditional acute effect that is not comparable with general-population estimates. Relevance: low-moderate - short-lag acute effects are what real-time LCS alerts are meant to act on.

doi: 10.1136/bmjopen-2026-120476

Respiratory & allergic — 1 record

Morales El Paso port-of-entry $PM_{2.5}$ immigration COPD census tracts 2026 J Immigr Minor Health

Breathing at the Border: $PM_{2.5}$ Exposure, Immigration, and COPD Across Census Tracts Close to Ports-of-Entry

El Paso census tracts; NASA satellite-derived $PM_{2.5}$ merged with CDC tract COPD prevalence and ACS social vulnerability; GLMs with distance to the port-of-entry. $PM_{2.5}$ and percent foreign-born non-citizens were positively associated with COPD prevalence; tracts within three miles of a port had higher prevalence, and the $PM_{2.5}$ effect was stratified by proximity. No coefficients or CIs in the abstract. Limitation: CDC PLACES prevalence is itself small-area modelled from BRFSS demographics, so regressing it on demographic covariates is partly circular; smoking is not addressed; satellite $PM_{2.5}$ at ~1 km cannot resolve truck-idling gradients at bridge queues. Relevance: high - port-of-entry diesel queues are a textbook case for dense LCS siting.

doi: 10.1007/s10903-026-02022-7

Reproductive & developmental — 1 record

Falcon-Gracia BiSC placental sncRNA air pollution maternal smoking 2026 medRxiv (preprint)

Impact of air pollution and tobacco smoke on the placental small non-coding RNA landscape

390 chorionic villus samples from the Barcelona Life Study Cohort, small-RNA sequencing; 979 sncRNA clusters (miRNA 33%, piRNA 26%, snoRNA 12%, tRF 8%). Maternal smoking was associated with miR-421 down-regulation and up-regulation of SNORD113/114 snoRNAs at the imprinted DLK1-DIO_3 locus and piR-26394. No sncRNA was associated with $PM_{2.5}$, BC or NO_2 after multiple-testing correction; several miRNAs were suggestive. Limitation: preprint; a null after FDR across ~1000 features at n=390 says little about effect absence, and pregnancy-average LUR exposure is a blunt contrast for trimester-specific placental biology. Relevance: low - a usefully reported null for PM against a positive smoking control.

doi: 10.64898/2026.10.16.26365732

Exposure assessment & modelling 6 records

Makacha PRECISE personal ambient $PM_{2.5}$ monitoring framework pregnancy SSA 2026 Front Glob Womens Health

An integrated climate-structured personal and ambient air pollution monitoring framework embedded within a pregnancy and postpartum cohort in Sub-Saharan Africa

PRECISE cohort: 343 pregnant or postpartum women (Gambia 160, Kenya 105, Mozambique 78), two 5-day wearable deployments in opposing seasons recording $PM_{2.5}$, T, RH and GPS/accelerometry; six fixed health-facility monitors. Calibration by repeated pre-deployment co-location with reference instruments and in-country Deming-regression harmonisation. Yield 3,190 person-days personal and 328 ambient days; inter-sensor R2 >0.72 pre-deployment and >0.75 in country; ambient capture >85% in Gambia and Kenya, lower in Mozambique (connectivity). Limitation: inter-sensor R2 is precision, not accuracy - the reference co-location bias and slope, and the RH correction applied, are not in the abstract; 5-day windows in a purposive subsample limit pregnancy-average inference. Relevance: high - a reusable LCS QA protocol for low-resource cohorts.

doi: 10.3389/fgwh.2026.1849628

Nolles IGP vs NCP $PM_{2.5}$ pollution forming efficiency CAMS 2026 Environ Sci Pollut Res Int

Indo-Gangetic Plain experiences higher aerosol pollution than North China Plain despite lower emissions

CAMS EAC4 reanalysis, CAMS-GLOB-ANT emissions and ERA5 for 2019. The NCP emitted 65% more primary PM and precursors (28,860 vs 17,464 kt) yet the IGP had higher annual $PM_{2.5}$ (123.6 vs 113.2 $\mu g\,m^{-3}$); NCP logged more bad-air days (251 vs 238) but the IGP longer episodes, up to 170 consecutive days around Delhi. A 'pollution forming efficiency' was nearly double in the IGP in winter, attributed to winds <6 km/h and PBL <300 m. Limitation: both concentration and emissions are model products for one year; EAC4 assimilates AOD and is biased in the IGP, and emission inventories under-count agricultural and residential burning there, which inflates the efficiency ratio. Relevance: moderate - quantifies what reanalysis cannot replace ground networks for.

doi: 10.1007/s11356-026-38305-7

Kirsanov COSMO-ART Saint Petersburg UHI pollution PM_{10} feedback 2026 Atmosphere

Assessing Urban Pollution and Heat Island Feedback in a Northern Megacity Using the Fully Coupled COSMO-ART Model

COSMO-ART with TERRA_URB for winter and summer episodes in Saint Petersburg. UHI-driven mixing and deeper PBL cut near-surface primary pollutants 15-54% (winter) and 5-33% (summer), PM_{10} peaks by >44%, while O_3 rose. Aerosol radiative feedback on the UHI was weak (hundredths of a degree, peaks <0.3 C), cooling in summer and warming over winter snow, under a fifth of canopy forcing. Limitation: two episodes, model-only; no comparison with observed PM_{10} is reported in the abstract, and PM_{10} in a snowy city is dominated by road-dust resuspension that emission inventories handle poorly. Relevance: moderate - urban/rural LCS gradients conflate emission density with UHI-enhanced dilution.

doi: 10.3390/atmos17101030

Imran Chinese cities urban air quality determinants interpretable ML 2014-2023 J Environ Manage

Spatio-temporal variation and determinants of urban air quality in Chinese cities during 2014-2023: an interpretable machine learning analysis

METADATA ONLY. Title and authors only; no abstract retrievable at harvest. The title describes a decade (2014-2023) of spatio-temporal variation in Chinese urban air quality with interpretable ML (SHAP-type attribution) of its determinants. Pollutants, number of cities, predictors and model skill cannot be stated. Limitation: unverifiable beyond the title; SHAP attributions on correlated socio-economic and meteorological predictors are not causal, and the 2014-2023 decline is dominated by the 2013 and 2018 action plans. Relevance: low - monitor-network secondary analysis.

doi: 10.1016/j.jenvman.2026.131184

Li atmospheric cadmium concentration deposition China modelling 2026 Atmos Environ

Modeling the spatial distributions and seasonal variations of atmospheric cadmium concentration and deposition in China

METADATA ONLY. Title and authors only; no abstract retrievable at harvest. National modelling of atmospheric cadmium - carried almost entirely on fine particles - with its spatial and seasonal distribution and deposition. Model, inventory, resolution and evaluation against measured Cd cannot be stated. Limitation: unverifiable beyond the title; metal inventories for smelting and coal are the dominant uncertainty. Relevance: low - mass-only LCS cannot see toxic trace-metal fractions, which is why such composition models remain needed.

doi: 10.1016/j.atmosenv.2026.122427

Liu Beijing autumn-winter $PM_{2.5}$ oxidative potential source predictors 2026 Atmos Environ

Autumn-winter variability and source-related predictors of $PM_{2.5}$ oxidative potential in urban Beijing

METADATA ONLY. Title and authors only; no abstract retrievable at harvest. An autumn-winter campaign relating $PM_{2.5}$ oxidative potential in urban Beijing to source-related predictors (heating, traffic, secondary formation presumably). Assay (DTT, AA, GSH), sample count and predictive model cannot be stated. Limitation: unverifiable beyond the title; a single season and site. Relevance: high - follows the 19 Oct Guangdong OP finding; if OP is predictable from source tracers, a subset of networked sensors with composition proxies could map it.

doi: 10.1016/j.atmosenv.2026.122428

Sensing, forecasting & instrumentation 5 records

Intra four co-deployed electrostatic PM monitors power plant agreement 2026 Research Square (preprint)

Multi-point monitoring of PM_{10}, $PM_{2.5}$, and $PM_1.0$ at a combined-cycle power plant: field agreement between four co-deployed electrostatic monitors

Four prototype three-channel electrostatic mass monitors at a combined-cycle plant, Feb-Jun 2020, 92-100% daily recovery; station means PM_{10} 44.9-54.1, $PM_{2.5}$ 24.8-30.8, $PM_1.0$ 7.5-11.9 $\mu g\,m^{-3}$. Fixed station differences explained 0.7% of log 24-h PM_{10} variance but the day-by-station interaction 57.1%; pairwise 95% limits of agreement spanned ratios 0.35-3.26 (PM_{10}) and 0.33-2.83 ($PM_{2.5}$); single-unit ICC 0.42 for PM_{10}, 0.75 for the four-unit mean. Disagreement did not shrink with averaging. Limitation: preprint; no reference instrument, so agreement says nothing about accuracy; units were not co-located side by side first, so unit drift and real micro-gradients are separated only by inference. Relevance: high - the same argument applies to any LCS network: station contrasts need routine co-location checks.

doi: 10.21203/rs.3.rs-10901106/v1

Li runway visual range model brown carbon aerosol chromatic effects 2026 Opt Lett

Refined runway visual range measurement model incorporating brown carbon aerosol effects

Optics Letter: conventional runway visual range (RVR) algorithms treat black carbon and scattering aerosol only; brown carbon's wavelength-selective absorption introduces two systematic errors - extinction spectral mismatch between the transmissometer/forward-scatter wavelength and the photopic band, and a chromatic shift in the visual threshold. A refined RVR model including BrC chromatic effects is proposed and 'experimentally validated'. Limitation: no error magnitudes, BrC loadings or validation data in the abstract. Relevance: moderate - the same single-wavelength assumption underlies optical PM sensors, whose mass conversion is biased by absorbing, spectrally selective aerosol.

doi: 10.1364/ol.611004

Wu ProNet physics-AI coupled CTM $PM_{2.5}$ O_3 forecasting 338 cities 2026 Natl Sci Rev

Fusing physics and AI enhances accuracy and diagnostic capabilities in air quality forecasting

ProNet embeds observation-constrained neural correctors inside the governing equations of a chemical transport model rather than post-processing its output. On 2022-23 data across 338 Chinese cities it beat the physical CTM and AI-only baselines, generalised better to extreme events outside training, and surpassed forecaster-adjusted operational forecasts by 6.1% ($PM_{2.5}$) and 3.0% (O_3). The correctors yield process-level error bookkeeping that exposes systematic CTM parameterisation biases. Limitation: '6.1% accuracy' is an unspecified metric; one two-year period invites temporal leakage, and process attribution from learned correctors is not unique - compensating errors can be assigned to the wrong process. Relevance: high - in-equation assimilation is the natural slot for dense calibrated LCS observations.

doi: 10.1093/nsr/nwag570

Sharma Singapore aerosol type classification ground remote sensing reanalysis 2026 Aerosol Air Qual Res

Aerosol Types Classification in the Tropical Climate of Singapore by Integrating Ground-Based Remote Sensing and Reanalysis Datasets

METADATA ONLY. Title and authors only; no abstract retrievable at harvest. Aerosol-type classification in tropical Singapore combining ground-based remote sensing (AERONET sun photometer presumably) with reanalysis (MERRA-2/CAMS). Classification scheme, period and the share of transboundary smoke cannot be stated. Limitation: unverifiable beyond the title; columnar typing does not map one-to-one onto surface PM. Relevance: moderate - aerosol type drives optical-LCS response (hygroscopicity, refractive index) and so the calibration regime a tropical network needs.

doi: 10.1007/s44408-026-00164-9

Kheder Foehn generative km-scale global $PM_{2.5}$ forecast 2026 arXiv (preprint)

Global probabilistic forecasting of air pollution

Foehn, a generative model trained on global kilometre-scale $PM_{2.5}$ estimates that also computes air-mass origin. In one year of retrospective forecasts it beat the operational forecast at three of four stations; its ensembles are reported as calibrated, and the ensemble mean detects local pollution extremes four days ahead about as well as the operational system does one day ahead. Limitation: preprint; 'kilometre-scale estimates' are themselves satellite/model fusion products, so skill is partly measured against the training distribution, and the authors concede real-time km-scale inputs do not yet exist - operational use is hypothetical. Skill metrics and station counts are not given in the abstract. Relevance: high - calibrated LCS networks are the plausible real-time km-scale input this model says it lacks.

doi: 10.48550/arXiv.2610.25941

Occupational & indoor 2 records

Arif Agra leather footwear facilities indoor $PM_{2.5}$ CO_2 formaldehyde 2026 Research Square (preprint)

Indoor Air Quality and Thermal Environment Across Three Leather Footwear Facilities in Agra India

Three leather-footwear facilities (cottage, small, organised), 27 months, 608 sessions. Facility dominated variance for every parameter (partial eta2 0.10-0.87); cottage vs organised means $PM_{2.5}$ 2.0-fold, CO_2 1.9-, formaldehyde 3.0- and VOC 3.5-fold higher. Season dominated PM and thermal variables, tracking ventilation; indoor/outdoor $PM_{2.5}$ ratios 0.48-1.00 across facilities; CO_2 above the ASHRAE threshold in 99.5% of cottage sessions vs 0% organised. Limitation: preprint; one facility per tier, so scale is confounded with building; instruments and their calibration are not stated, and spot sessions are not shift-weighted personal exposure. Relevance: moderate - informal-sector workplaces are where portable LCS are the only feasible monitor.

doi: 10.21203/rs.3.rs-10962722/v1

Kim pig building air cleaner TSP $PM_{2.5}$ productivity 2026 J Anim Sci Technol

Association between indoor air pollutants and pig productivity index by on-site application of air cleaner

One commercial pig building, control vs air-cleaner-treated room, monitored six months across rearing stages. Air-cleaner reduction efficiency rose with stage, 5-10% at farrowing to 30-45% at fattening; treated rooms showed better weight gain, feed conversion, activity index and mortality. Weight gain and activity correlated with TSP and $PM_{2.5}$ (and airborne fungi). Limitation: one room per arm, no randomisation or replication - room effects are inseparable from treatment; the cleaner technology and PM instrumentation are not stated. Relevance: low - livestock housing is a large agricultural PM source and an occupational exposure for farm workers, but this design supports neither inference.

doi: 10.5187/jast.2500087

Burden, policy & mitigation 1 record

Prieto Perez Huszar future central European air quality emission changes 2026 Atmos Environ

The role that changes in emissions play in future central European air-quality

METADATA ONLY. Title and authors only (Prieto Perez, Huszar, Karlicky); no abstract retrievable at harvest. The group runs regional climate-chemistry (RegCM/WRF-CAMx) scenarios; the title separates the emission-change contribution to future central European air quality from climate change. Pollutants, scenarios and horizons cannot be stated. Limitation: unverifiable beyond the title; scenario attribution depends on the emission pathway chosen. Relevance: low - policy context for the PM levels European LCS networks will measure.

doi: 10.1016/j.atmosenv.2026.122404

Other clinical endpoints 1 record

Wen landscape fire $PM_{2.5}$ renal hospitalisations six-country time-series 2026 Environ Sci Technol

Landscape Fire-Sourced $PM_{2.5}$ and Cause-Specific Renal Hospitalizations: A Multicountry Two-Stage Time-Series Analysis of 12.5 Million Admissions

12.5 million renal admissions across 1020 locations in six countries, 2000-2019; two-stage time-series with fire-sourced $PM_{2.5}$ separated from the non-fire remainder. Per 10 $\mu g\,m^{-3}$ cumulative lag 0-5, RR 1.011 (1.007-1.015) for all renal admissions, strongest for acute kidney injury; associations larger for fire-sourced than non-fire $PM_{2.5}$, larger in warmer areas and graded by GDP per capita. Attributable burden 1.38 admissions per 100,000 per year. Limitation: the fire/non-fire split comes from a chemical-transport-plus-ML partition, not measured tracers, so 'fire PM is more toxic' partly inherits the partition's error structure; heat co-occurs with smoke and dehydration drives AKI, and residual confounding by temperature is the obvious threat. Relevance: moderate - smoke-episode exposure is where LCS density most improves over sparse FRM/FEM coverage.

doi: 10.1021/acs.est.6c04677

Corpus register

First author	Focus	Journal	Design	Metric	Region
Cardiovascular & metabolic					
Huang $PM_{2.5}$ coronary artery dilatation Kawasaki disease case-crossover	Association between PM(2.5) and coronary artery dilatation with Kawasaki disease in China	*Eur J Pediatr*	Case-crossover	Short-term $PM_{2.5}$ (distributed lag 0-7 d)	China (single paediatric centre)
Wongcharoen short-term $PM_{2.5}$ MACE high-risk population nationwide case-crossover	Associations between short-term PM(2.5) exposure and major cardiovascular events in high-risk population: a nationwide case-crossover study	*BMJ Open*	Metadata only (no abstract)	Short-term $PM_{2.5}$	Thailand (inferred from author group; not stated in record)
Reproductive & developmental					
Falcon-Gracia BiSC placental sncRNA air pollution maternal smoking	Impact of air pollution and tobacco smoke on the placental small non-coding RNA landscape	*medRxiv (preprint)*	Cross-sectional (birth cohort)	Prenatal $PM_{2.5}$, black carbon, NO2 (hybrid dispersion-LUR)	Spain (Barcelona, BiSC)
Respiratory & allergic					
Morales El Paso port-of-entry $PM_{2.5}$ immigration COPD census tracts	Breathing at the Border: $PM_{2.5}$ Exposure, Immigration, and COPD Across Census Tracts Close to Ports-of-Entry	*J Immigr Minor Health*	Ecological / registry analysis	Census-tract $PM_{2.5}$ (NASA satellite-derived)	USA (El Paso, Texas, US-Mexico border)
Occupational & indoor					
Arif Agra leather footwear facilities indoor $PM_{2.5}$ CO2 formaldehyde	Indoor Air Quality and Thermal Environment Across Three Leather Footwear Facilities in Agra India	*Research Square (preprint)*	Instrument evaluation	Indoor $PM_{2.5}$/PM_{10}, I/O $PM_{2.5}$ ratio, CO2, formaldehyde, TVOC	India (Agra)
Kim pig building air cleaner TSP $PM_{2.5}$ productivity	Association between indoor air pollutants and pig productivity index by on-site application of air cleaner	*J Anim Sci Technol*	Intervention field trial (non-randomised)	TSP, $PM_{2.5}$, airborne fungi in pig housing	South Korea (commercial pig farm)
Sensing, forecasting & instrumentation					
Intra four co-deployed electrostatic PM monitors power plant agreement	Multi-point monitoring of PM_{10}, $PM_{2.5}$, and PM1.0 at a combined-cycle power plant: field agreement between four co-deployed electrostatic monitors	*Research Square (preprint)*	Instrument evaluation	PM_{10}, $PM_{2.5}$, PM1.0 (corona charging, impaction, Faraday-cup electrometry)	Thailand (Samut Prakan)
Kheder Foehn generative km-scale global $PM_{2.5}$ forecast	Global probabilistic forecasting of air pollution	*arXiv (preprint)*	Deep-learning forecasting	Global kilometre-scale $PM_{2.5}$ estimates (training and target)	Global
Li runway visual range model brown carbon aerosol chromatic effects	Refined runway visual range measurement model incorporating brown carbon aerosol effects	*Opt Lett*	Theoretical / optical modelling	Brown carbon aerosol spectral extinction	Not stated in record
Sharma Singapore aerosol type classification ground remote sensing reanalysis	Aerosol Types Classification in the Tropical Climate of Singapore by Integrating Ground-Based Remote Sensing and Reanalysis Datasets	*Aerosol Air Qual Res*	Metadata only (no abstract)	Columnar aerosol optical properties (AOD, Angstrom exponent presumably)	Singapore
Wu ProNet physics-AI coupled CTM $PM_{2.5}$ O3 forecasting 338 cities	Fusing physics and AI enhances accuracy and diagnostic capabilities in air quality forecasting	*Natl Sci Rev*	Deep-learning forecasting	$PM_{2.5}$ and O3 forecasts (338 Chinese cities)	China (338 cities)
Exposure assessment & modelling					
Imran Chinese cities urban air quality determinants interpretable ML 2014-2023	Spatio-temporal variation and determinants of urban air quality in Chinese cities during 2014-2023: an interpretable machine learning analysis	*J Environ Manage*	Metadata only (no abstract)	City-level air quality ($PM_{2.5}$ presumably among pollutants)	China (multi-city)
Kirsanov COSMO-ART Saint Petersburg UHI pollution PM_{10} feedback	Assessing Urban Pollution and Heat Island Feedback in a Northern Megacity Using the Fully Coupled COSMO-ART Model	*Atmosphere*	Chemical-transport modelling	Near-surface PM_{10}, NO, CO, O3; aerosol radiative feedback	Russia (Saint Petersburg)
Li atmospheric cadmium concentration deposition China modelling	Modeling the spatial distributions and seasonal variations of atmospheric cadmium concentration and deposition in China	*Atmos Environ*	Metadata only (no abstract)	Particle-bound atmospheric cadmium (concentration and deposition)	China (national)
Liu Beijing autumn-winter $PM_{2.5}$ oxidative potential source predictors	Autumn-winter variability and source-related predictors of $PM_{2.5}$ oxidative potential in urban Beijing	*Atmos Environ*	Metadata only (no abstract)	$PM_{2.5}$ oxidative potential (assay not stated) and source tracers	China (Beijing)

First author	Focus	Journal	Design	Metric	Region
Makacha PRECISE personal ambient $PM_{2.5}$ monitoring framework pregnancy SSA	An integrated climate-structured personal and ambient air pollution monitoring framework embedded within a pregnancy and postpartum cohort in Sub-Saharan Africa	*Front Glob Womens Health*	Instrument evaluation	Minute-resolved personal and fixed-site $PM_{2.5}$ (wearable LCS)	Sub-Saharan Africa (Gambia, Kenya, Mozambique)
Nolles IGP vs NCP $PM_{2.5}$ pollution forming efficiency CAMS	Indo-Gangetic Plain experiences higher aerosol pollution than North China Plain despite lower emissions	*Environ Sci Pollut Res Int*	Reanalysis / inventory analysis	CAMS EAC4 $PM_{2.5}$ vs CAMS-GLOB-ANT emissions	South Asia and China (IGP vs NCP)
Burden, policy & mitigation					
Prieto Perez Huszar future central European air quality emission changes	The role that changes in emissions play in future central European air-quality	*Atmos Environ*	Metadata only (no abstract)	Future PM and O3 under emission scenarios (presumably)	Central Europe
Other clinical endpoints					
Wen landscape fire $PM_{2.5}$ renal hospitalisations six-country time-series	Landscape Fire-Sourced $PM_{2.5}$ and Cause-Specific Renal Hospitalizations: A Multicountry Two-Stage Time-Series Analysis of 12.5 Million Admissions	*Environ Sci Technol*	Multi-location time-series	Landscape-fire-sourced vs non-fire $PM_{2.5}$ (per 10 $\mu g/m^3$, lag 0-5)	Multi-country (Australia, Brazil, Canada, Chile, New Zealand, Thailand)

Provenance and method

Entry window **20 October 2026**, single day (Tuesday). Harvester legs: **PubMed** health **20**, sensing **4**; **Europe PMC 3** (index lag; 0 AGRICOLA); **Crossref by ISSN 52** (2 ES&T front-matter dropped); **arXiv** no entry dated 20 October; **OpenAlex** HTTP 429. 76 unique, 5 already carried, 71 fresh. **PubMed connector** (EDAT 20 Oct) 18 PMIDs, 5 appended — **76** candidates screened, **18 admitted**, **58 rejected** with logged reasons, plus **1 held** record released (arXiv 2610.25941), giving **19**. **Consensus** calibration sweep: 10 hits, 0 new. **DOI gate** 0 fail / 1 warn (arXiv DataCite DOI live at doi.org, not in Crossref). **Trial watch**: 0 registrations updated 20–21 October. Abstract retry (PubMed, Europe PMC, Semantic Scholar) left **6 carried on metadata alone**. **Scholar Gateway** not called. Abstracts remain the copyright of their publishers.